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Reciprocating tool

Abstract

A reciprocating tool includes a motor, a motion converting mechanism, a body housing, an inner housing, a support body, and a counterweight. The motion converting mechanism is operably coupled to a motor shaft and is configured to convert rotation into a linear reciprocating motion along a driving axis. The inner housing is within the body housing and houses at least a portion of the motion converting mechanism. The support body is originally separate from the inner housing and is attached to the inner housing in the body housing. The counterweight is operably coupled to the motion converting mechanism and is configured to be driven by the motion converting mechanism. The counterweight is supported by the support body to be pivotable around a pivot axis that extends in a direction orthogonal to the driving axis. At least a portion of the counterweight is housed in the inner housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to Japanese patent applications No. 2022-177511 filed on Nov. 4, 2022, No. 2023-104865 filed on Jun. 27, 2023, and No. 2023-104868 filed on Jun. 27, 2023. The contents of the foregoing applications are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a reciprocating tool that is configured to linearly reciprocate a tool accessory.

BACKGROUND

(3) A reciprocating tool is configured to linearly reciprocate a tool accessory along a driving axis. When the tool accessory is driven, relatively large vibration is caused in an extension direction of the driving axis. Thus, some known reciprocating tools include a counterweight for reducing the vibration. For example, a power tool having a hammer mechanism that is disclosed in Japanese laid-open patent publication No. 2013-013951 includes a motion converting mechanism and a counter weight. The motion converting mechanism is configured to be driven by a motor. The counterweight is disposed in an internal space of a housing member (an inner housing) and is supported by the housing member to be pivotable around a pivot axis. The counterweight is operably coupled to the motion converting mechanism and is pivoted around the pivot axis by the motion converting mechanism.

SUMMARY

(4) The above-described housing member needs to support the counterweight that is operably coupled to the motion converting mechanism. Therefore, there are constraints in design of the housing member. Thus, there is still room for improvement in a supporting structure of the counterweight in view of the degree of freedom in the design of the housing member.
(5) Accordingly, it is a non-limiting object of the present disclosure to provide a reciprocating tool having an improved supporting structure of a counterweight.
(6) One non-limiting embodiment according to the present disclosure herein provides a reciprocating tool that is configured to linearly reciprocate a tool accessory. The reciprocating tool

includes a motor, a motion converting mechanism, a body housing, an inner housing, a support body, and a counterweight.

(7) The motor has a motor shaft that is rotatable round a motor axis. The motion converting mechanism is operably coupled to the motor shaft. The motion converting mechanism is configured to convert rotation into linear reciprocating motion along a driving axis. The driving axis defines a front-rear direction of the reciprocating tool. The inner housing is disposed within the body housing. The inner housing houses at least a portion of the motion converting mechanism. The support body is originally separate (discrete) from the inner housing, and is attached to the inner housing. The counterweight is operably coupled to the motion converting mechanism. The counterweight is configured to be driven by the motion converting mechanism. The counterweight is supported by the support body to be pivotable around a pivot axis that extends in a direction orthogonal to the driving axis. At least a portion of the counterweight is within the inner housing.

(8) The reciprocating tool of this embodiment includes the counterweight that is driven by the motion converting mechanism. Thus, the counterweight can effectively reduce vibration generated during reciprocating driving of the tool accessory. Further, the counterweight is supported by the support body that is originally separate from the inner housing (i.e., a component (part) that is separate (discrete) from the inner housing) and is supported by the inner housing. Therefore, a supporting structure for the counterweight does not need to be formed integrally with the inner housing. Consequently, the degree of freedom in design of the inner housing can be enhanced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a rotary hammer.

(2) FIG. 2 is a sectional view of the rotary hammer.

(3) FIG. 3 is a partial, enlarged view of FIG. 2.

(4) FIG. 4 is a perspective view of an inner housing.

(5) FIG. 5 is a perspective view of a support body and a driving mechanism that are mounted on the inner housing.

(6) FIG. 6 is another partial, enlarged view of FIG. 2.

(7) FIG. 7 is a sectional view taken along line VII-VII in FIG. 3.

(8) FIG. 8 is a perspective view of a counterweight that is supported by the support body.

(9) FIG. 9 is a perspective view of the driving mechanism to which the support body and the counterweight are mounted.

(10) FIG. 10 is yet another partial, enlarged view of FIG. 2.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(11) In one non-limiting embodiment according to the present disclosure, the counterweight may be supported by the support body via a support shaft that extends along the pivot axis. The support shaft may be in an internal space of the inner housing. The internal space of the inner housing herein means a space that is surrounded by an outer shell (a wall part or wall parts) of the inner housing. According to this embodiment, at least a portion of the counterweight that is supported by the support body via the support shaft can be easily placed within the inner housing.

(12) In addition or in the alternative to the preceding embodiments, the support body may include a main body and an arm part. The main body is at least partially on an outside of the inner housing. The arm part protrudes from the main body into the inner housing and supports the support shaft. According to this embodiment, the support shaft that is supported by the arm part can be easily placed within the inner housing.

(13) In addition or in the alternative to the preceding embodiments, the reciprocating tool may further include a seal member that is disposed (interposed) between the body housing and the inner

housing. The seal member may be frontward of the pivot axis of the counterweight in the front-rear direction. In a known-structure in which the inner housing itself directly supports the counterweight, in order to lubricate a supporting structure of the counterweight, the seal member is disposed rearward of the pivot axis of the counterweight. On the contrary, in this embodiment, the seal member is disposed frontward of the pivot axis. Owing to such configuration, a surface area of a portion of the inner housing that extends rearward of the seal member can be made larger. Consequently, the inner housing can be effectively cooled.

(14) In addition or in the alternative to the preceding embodiments, an extension direction of the pivot axis may define a left-right direction of the reciprocating tool. A direction that is orthogonal to the front-rear direction and the left-right direction may define an up-down direction of the reciprocating tool. The body housing may include an air inlet opening and an air outlet opening. The air inlet opening may be above an upper end of the seal member in the up-down direction, and the air outlet opening may be below a lower end of the seal member in the up-down direction. According to this embodiment, the inner housing can be effectively cooled by air that flows into the body housing through the air inlet opening and flows out of the body housing through the air outlet opening.

(15) In addition or in the alternative to the preceding embodiments, the reciprocating tool may further include a spindle and a hammer (striking) element that is disposed within the spindle. The spindle may be configured to hold the tool accessory to be movable along the driving axis. The striker may be configured to be driven by the motion converting mechanism to apply a striking force to the tool accessory. In other words, the reciprocating tool may be configured as a power tool having a hammer (striking) mechanism. The support body may be configured to support the spindle. According to this embodiment, the support body, which pivotably supports the counterweight, can be utilized to support the spindle as well.

(16) In addition or in the alternative to the preceding embodiments, the reciprocating tool may further include an intermediate shaft. The intermediate shaft may be operably coupled to the motor shaft and rotatable around a rotational axis that extends in parallel to the driving axis. A portion of the motion converting mechanism may be disposed on the intermediate shaft. According to this embodiment, a compact reciprocating tool having the counterweight can be achieved.

(17) In addition or in the alternative to the preceding embodiments, the motor axis may intersect the rotational axis of the intermediate shaft. According to this embodiment, a reciprocating tool having an L-shape can be achieved.

(18) In addition or in the alternative to the preceding embodiments, an extension direction of the pivot axis may define a left-right direction of the reciprocating tool. A direction that is orthogonal to the front-rear direction and the left-right direction may define an up-down direction of the reciprocating tool. The body housing may include an air inlet opening and an air outlet opening. The air inlet opening may be above the driving axis in the up-down direction. According to this embodiment, the inner housing can be effectively cooled by air that flows into the body housing through the air inlet opening and flows out of the body housing through the air outlet opening.

(19) In addition or in the alternative to the preceding embodiments, an air passage may be defined in the body housing. The air passage may be configured such that air, which has flowed into the body housing through the air inlet opening, flows along the inner housing, passes through an inside of the motor, and flows out of the body housing through the air outlet opening. According to this embodiment, the inner housing and the motor can be effectively cooled by the air that flows into the body housing through the air inlet opening and flows out of the body housing through the air outlet opening.

(20) A rotary hammer (hammer drill) **1** according to a representative, non-limiting embodiment of the present disclosure is now described with reference to FIGS. **1** to **10**. The rotary hammer **1** is a power tool that is configured to perform a hammer action. In the hammer action, a tool accessory **91** that is removably held by the rotary hammer **1** is hammered (stricken), and thereby the tool

accessory **91** is linearly reciprocated along a driving axis DX. Thus, the rotary hammer **1** is an example of a reciprocating tool and is also an example of a power tool having a hammer (striking) mechanism. The rotary hammer **1** may perform a rotary action and the hammer action at the same time, or perform the rotary action in the alternative to the hammer action. In the rotary action, the tool accessory **91** is rotationally driven around the driving axis DX.

(21) First, the general structure of the rotary hammer **1** is described. As shown in FIGS. **1** and **2**, an outline of the rotary hammer **1** is defined by a body housing **10** and a handle **17** that is coupled to the body housing **10**.

(22) The body housing **10** is a hollow body that is also referred to as a tool body or an outer shell. The body housing **10** of this embodiment includes a first housing part **11** and a second housing part **15**.

(23) As shown in FIG. **2**, the first housing part **11** mainly houses a spindle **40**, which is a member for holding a tool accessory (i.e., a tool accessory holding member), and a driving mechanism **4** for driving the tool accessory **91**. The spindle **40** is an elongate tubular member. One end portion of the spindle **40** in its longitudinal direction is structured as a tool holder **41** that is configured to removably hold the tool accessory **91**. A longitudinal axis of the tool holder **41** (the spindle **40**) defines the driving axis DX of the tool accessory **91**.

(24) The first housing part **11** extends along the driving axis DX. A first end portion of the first housing part **11** in an extension direction of the driving axis DX has a hollow cylindrical shape, and the tool holder **41** is housed in this end portion, which is also referred to as a barrel part **111**. The remaining portion of the first housing part **11** has a tubular shape that has a larger diameter than the barrel part **111** and has an opening **113** at an end that is opposite to the barrel part **111**. A metal inner housing **20** is fitted into a second end portion defining the opening **113** of the first housing part **11** so as to close the opening **113**. Accordingly, a first space **101** is defined that is surrounded by the inner housing **20** and the first housing part **11**. The spindle **40** and the driving mechanism **4** are disposed in the first space **101**. Thus, an entirety of the first housing part **11** and the inner housing **20** may be defined as one single housing (a so-called gear housing) that houses the spindle **40** and the driving mechanism **4**.

(25) The second housing part **15** mainly houses a motor **31**. The second housing part **15** is coupled to the second end portion of the first housing part **11** that is opposite to the barrel part **111** in the extension direction of the driving axis DX, and extends in a direction that intersects (more specifically, substantially orthogonal to) the driving axis DX. The body housing **10** including the first housing part **11** and the second housing part **15** is thus formed in an L-shape as a whole. A portion of the second housing part **15** surrounds or covers the inner housing **20** from the outside thereof.

(26) The handle **17** is a hollow member that has a U-shape as a whole. Opposite ends of the handle **17** are coupled to the body housing **10** (the second housing part **15**). The handle **17** includes a grip part **171** that is configured to be gripped by a user. The grip part **171** extends in a direction that intersects (specifically, substantially orthogonal to) the driving axis DX. The grip part **171** extends generally in parallel to an extension direction of the second housing part **15**. The grip part **171** has a trigger **173** to be depressed by the user, and houses a switch **175**. A lower end portion of the handle **17** houses a controller **30** that is configured to control operation of the rotary hammer **1**. A battery **93** is removably mounted to the lower end portion of the handle **17**. When the trigger **173** is depressed and thus the switch **175** is turned ON, the controller **30** starts driving of the motor **31**, so that the driving mechanism **4** drives the tool accessory **91**.

(27) The detailed structure of the rotary hammer **1** is now described. In the following description, for the sake of convenience, the extension direction of the driving axis DX is defined as a front-rear direction of the rotary hammer **1**. In the front-rear direction, the side on which a distal end of the tool holder **41** is located (i.e., the side on which the tool accessory **91** is inserted into the tool holder **41**) is defined as a front side of the rotary hammer **1**, while the opposite side is defined as a rear

side of the rotary hammer **1**. A direction that is orthogonal to the driving axis DX and that substantially corresponds to an extension direction of the second housing part **15** (also an extension direction of the grip part **171**) is defined as an up-down direction of the rotary hammer **1**. In the up-down direction, the side on which the first housing part **11** is located is defined as an upper side of the rotary hammer **1**, while an opposite side is defined as a lower side of the rotary hammer **1**. A direction that is orthogonal to both of the front-rear direction and the up-down direction is defined as a left-right direction.

(28) First, the detailed structures of the body housing **10** and the inner housing **20** are described.

(29) As described above, the first housing part **11** is a tubular member as a whole, and the opening **113** at a rear end portion is closed by the inner housing **20**. As shown in FIGS. **3** to **5**, the inner housing **20** is a hollow body that has an opening **201** and an internal space **200**. The inner housing **20** is disposed such that the opening **201** is directed forward, and fitted into the rear end portion of the first housing part **11** that defines the opening **113**. More specifically, the inner housing **20** includes a tubular peripheral (circumferential) wall part **202** that defines the opening **201**, and a rear wall part **203** that is connected to the peripheral wall part **202**. The internal space **200** opens frontward, is surrounded by the peripheral wall part **202** in a circumferential direction around the driving axis DX, and a rear end of the internal space **200** is closed by the rear wall part **203**.

(30) An annular groove **204** is formed on an annular region, which is adjacent to the opening **201** and extends in the circumferential direction, of an outer surface of the peripheral wall part **202**. An annular seal member **28** is fitted in the groove **204** of the inner housing **20**. The seal member **28** is formed as, for example, an annular elastic member (e.g., a rubber O-ring). The seal member **28** is compressed between an inner surface of the rear end portion of the first housing part **11** and the outer surface of the peripheral wall part **202** of the inner housing **20**, so as to seal a gap between the first housing part **11** and the inner housing **20**. Thus, the closed first space **101** is defined by the first housing part **11** and the inner housing **20**. The spindle **40**, and the driving mechanism **4** are disposed in the first space **101**, together with lubricant (e.g., grease).

(31) As shown in FIGS. **1** and **2**, the second housing part **15** is a hollow body that is coupled to the rear end portion of the first housing part **11** and extends in the up-down direction. An upper half **151** of the second housing part **15** is coupled to the rear end portion of the first housing part **11**, using screws. In this embodiment, the second housing part **15** is formed by two halves that are originally divided in the left-right direction (i.e., left and right halves) and coupled to each other using screws. The inner housing **20** is within the upper half **151** of the second housing part **15**. The motor **31** is disposed within a lower half **153** of the second housing part **15** (a portion of the second housing part **15** that extends downward of a portion that houses the inner housing **20**).

(32) Owing to the above-described configuration, two spaces that are partitioned by the inner housing **20** are defined within the body housing **15**. More specifically, the two spaces include the first space **101** in which the spindle **40** and the driving mechanism **4** are disposed together with lubricant, and a second space **102** in which the motor **31** is mainly disposed. The first space **101** and the second space **102** are also separated (isolated) by the seal member **28** that is disposed (interposed) between the inner housing **20** and the body housing **10** (the first housing part **11**).

(33) The elements (structures) disposed within the body housing **10** are now described.

(34) First, the motor **31** is described. As shown in FIG. **2**, the motor **31** is within the lower half **153** of the second housing part **15**. The motor **31** has a motor body **310**, which includes a stator and a rotor, and a motor shaft **315**. The motor shaft **315** extends from the rotor and is configured to rotate integrally with the rotor around a motor axis MX. In this embodiment, the motor **31** is disposed such that the motor axis MX extends slightly obliquely relative to the up-down direction of the rotary hammer **1** and obliquely intersects the driving axis DX. However, the motor **31** may be disposed such that the motor axis MX extends in the up-down direction so as to orthogonally intersect the driving axis DX.

(35) A first bevel gear **33** is coupled to an upper end portion of the motor shaft **315**. The first bevel

gear **33** is configured to be rotated integrally with the motor shaft **315**. The motor shaft **315** is rotatably supported by two bearings **321** and **323**. The upper bearing **321** is supported by a lower rear end portion of the inner housing **20**. The bearing **321** rotatably supports the upper end portion of the motor shaft **315**. A teeth part of the first bevel gear **33** is disposed within the inner housing **20**. The teeth part herein refers to a portion of the first bevel gear **33** on which gear teeth **331** (see FIG. 3) are formed. The lower bearing **323** is supported by the second housing part **15** so as to rotatably support the motor shaft **315** below the motor body **310**.

(36) A fan **35** for cooling the motor **31** is fixed to a lower end portion of the motor shaft **315** (a portion extending downward from the bearing **323**). The fan **35** is configured to be rotated integrally with the motor shaft **315** so as to generate an air flow (cooling air) that flows into the body housing **10** through air inlet openings **105**, passes through the motor **31**, and flows out of the body housing **10** through air outlet openings **107**.

(37) As shown in FIGS. 1 and 2, in this embodiment, the air inlet openings **105** are formed in the upper end portion (through the upper wall part) of the second housing part **15**. More specifically, the air inlet openings **105** are rearward of the seal member **28** and frontward of the rear end of the inner housing **20** in the front-rear direction. Thus, the air inlet openings **105** are directly above the inner housing **20** in the up-down direction. The air outlet openings **107** are formed in the lower end portion of the second housing part **15**. More specifically, the air outlet openings **107** are provided at a left side part and a right side part of the lower end portion of the second housing part **15**.

(38) Owing to the air inlet openings **105** and the air outlet openings **107** that are thus arranged, an air passage is defined in the second space **102**. Specifically, the air is sucked in response to rotational driving of the fan **35** into the upper half **151** of the second housing part **15** through the air inlet opening **105**, and flows downward along the inner housing **20** between the inner surface of the second housing part **15** and the outer surface of the inner housing **20**. Further, the air passes between the stator and the rotor of the motor **31** disposed in the lower half **153** and reaches the fan **35**. The air is delivered from the fan **35** and flows out of the second housing **15** through the air outlet opening **107**.

(39) In this embodiment, as shown in FIG. 3, guide ribs **106** are provided adjacent to the air inlet openings **105** in the second housing part **15**. The guide ribs **106** are each configured to lead the air that flows into the second housing part **15** through the air inlet opening **105** toward the front side. Specifically, a lower portion of each guide rib **106** is inclined frontward and downward. Thus, the air that flows through the air inlet openings **105** is guided toward a portion of the inner housing **20** on which the seal member **28** is mounted.

(40) The spindle **40** is now described.

(41) As shown in FIG. 3, the spindle **40** of this embodiment is an elongate, stepped hollow cylindrical member. The spindle **40** is supported in an upper half of the first space **101** and extends in the front-rear direction.

(42) A front half of the spindle **40** defines the tool holder **41**. The tool accessory **91** is inserted into an opening **410** of the tool holder **41** at its front end such that the longitudinal axis of the tool accessory **91** coincides with the driving axis DX. When the tool accessory **91** is held by the tool holder **41**, the tool accessory **91** is allowed to move in the axial direction relative to the tool holder **41** and restricted from rotating around the longitudinal axis relative to the tool holder **41**. A rear half of the spindle **40** defines a cylinder **42** that slidably holds a piston **57**. In this embodiment, the spindle **40** is a single component including the tool holder **41** and the cylinder **42** that are integrally formed. However, the spindle **40** may be formed by coupling a plurality of separate (discrete) members.

(43) The spindle **40** is supported by two bearings **401** and **402** to be rotatable around the driving axis DX. The front bearing **401** is fitted into and supported by the front end portion (the barrel part **111**) of the first housing part **11** and rotatably supports the front end portion (the tool holder **41**) of the spindle **40**. The rear bearing **402** is supported by a support body **22** and rotatably supports the

rear end portion (the cylinder **42**) of the spindle **40**.

(44) As shown in FIG. 5, the support body **22** is an annular (ring-like) member as a whole. The support body **22** is made of metal. As shown in FIG. 3, the bearing **402** is fitted into the support body **22**. The support body **22** is coupled (attached, fixed) to the front end portion of the upper half of the inner housing **20** and is fitted into the upper portion of the first housing part **11**. With this structure, the rear end portion of the spindle **40** is supported by the first housing part **11** via the bearing **402** and the support body **22**.

(45) The driving mechanism **4** is now described.

(46) As shown in FIG. 3, the driving mechanism **4** is operably coupled to the motor **31** (the motor shaft **315**). The driving mechanism **4** is configured to be driven by the power of the motor **31**. The driving mechanism **4** of this embodiment includes a hammer (striking) mechanism **5** that is configured to perform the hammer action and a rotation transmitting mechanism **6** that is configured to perform the rotary action. The hammer mechanism **5** includes a motion converting mechanism **51** and a hammer (striking) element **58**.

(47) As shown in FIGS. 3 and 6, the motion converting mechanism **51** is operably coupled to the motor shaft **315**. The motion converting mechanism **51** is configured to convert rotation of the motor shaft **315** into linear motion along the driving axis DX (specifically, linear motion of the piston **57**) for driving the tool accessory **91**. In this embodiment, the motion converting mechanism **51** is of an oscillating type (pivot type). The motion converting mechanism **51** includes a rotary member **54** and an oscillating member **55** that are disposed on (around) an intermediate shaft **52**, and the piston **57** that is disposed within the spindle **40** (the cylinder **42**).

(48) The intermediate shaft **52** is disposed in the lower half of the first space **101**. The intermediate shaft **52** is supported to be rotatable around a rotational axis RX that is parallel to the driving axis DX. The rotational axis RX thus extends in the front-rear direction. A second bevel gear **53** is fixedly coupled to a rear end portion of the intermediate shaft **52**. The second bevel gear **53** is configured to be rotated integrally with the intermediate shaft **52**.

(49) The second bevel gear **53** includes a cylindrical mount part **533** and a teeth part having gear teeth **531**. The mount part **533** is fixed around (onto an outer periphery of) the rear end portion of the intermediate shaft **52**. A rear end of the second bevel gear **53** (a rear end of the teeth part) is rearward of the rear end of the intermediate shaft **52**. Thus, a portion of the second bevel gear **53** protrudes rearward from the rear end of the intermediate shaft **52**. The second bevel gear **53** meshes with the first bevel gear **33** that is fixed on the motor shaft **315**, and configured to be rotated integrally with the intermediate shaft **52** when the motor shaft **315** is rotated. The first bevel gear **33** and the second bevel gear **53** form a gear speed reducer. Thus, the gear teeth **531** of the second bevel gear **53** has a diameter that is larger than that of the gear teeth **331** of the first bevel gear **33**.

(50) The intermediate shaft **52** is rotatably supported by two bearings **521** and **522**. The front bearing **521** is fitted around the front end portion of the intermediate shaft **52**. The bearing **521** is supported by the front end portion of the body housing **10** (the first housing part **11**). The rear bearing **522** is fitted around the mount part **533** of the second bevel gear **53**. The bearing **522** is supported by a bearing support **23** that is fixed to the inner housing **20**. Thus, in this embodiment, the rear end portion of the intermediate shaft **52** is rotatably supported via the second bevel gear **53**. Owing to such a structure, the intermediate shaft **52** can be made shorter, compared to a known structure in which a portion of the intermediate shaft **52** is directly supported by a bearing.

(51) The bearing support **23** is originally separate (discrete) from the inner housing **20**. The bearing support **23** is disposed within the inner housing **20** and is fixed to the inner housing **20** using screws. The bearing support **23** is configured to cover a peripheral portion of the teeth part and a front end portion of the second bevel gear **53**. The second bevel gear **53** is housed in a space that is defined between the rear wall part **203** of the inner housing **20** and the bearing support **23**. The bearing support **23** has openings **230** that are formed radially outward of the gear teeth **531** of the second bevel gear **53**. The openings **230** each communicate with an outer space of the bearing

support **23** and an internal space of the bearing support **23** in which the second bevel gear **53** is disposed.

(52) The rotary member **54** is disposed on (around) the intermediate shaft **52** and is configured to be rotate integrally with the intermediate shaft **52**. Although not described in detail because it is a well-known structure, a state of the rotary member **54** is changed between a first state and a second state by a mode changing mechanism according to a selected action mode. In the first state, the rotary member **54** rotates integrally with the intermediate shaft **52**. In the second state, the rotary member **54** idly rotates relative to the intermediate shaft **52**. The rotary member **54** rotates integrally with the intermediate shaft **52** in the first state only when the action mode for performing the hammer action is selected. The oscillating member **55** is operably coupled to the rotary member **54**. The oscillating member **55** is configured to be oscillated (pivoted) while the rotary member **54** rotates. The rotary member **54** and the oscillating member **55** are well-known components that may also be collectively called a swash bearing or a wobble bearing.

(53) The oscillating member **55** of this embodiment includes an annular ring part **550**, an oscillating arm **551** and a protrusion **552**. The ring part **550** is mounted around the rotary member **54** via rolling elements (e.g., balls). The oscillating arm **551** and the protrusion **552** each protrude radially outward from the ring part **550**. More specifically, the oscillating arm **551** protrudes upward from the ring part **550** in the up-down direction and is configured to be oscillated in the front-rear direction while the rotary member **54** rotates. The oscillating arm **551** is operably coupled to the piston **57**. The protrusion **552** is coaxial with (extends along) a longitudinal axis of the oscillating arm **551** and protrudes from the ring part **550** in a direction opposite to the oscillating arm **551**. Thus, the protrusion **552** is oscillated opposite to the oscillating arm **551** in the front-rear direction while the rotary member **54** rotates. The protrusion **552** is coupled to a counterweight **7**. The counterweight **7** and a supporting structure thereof will be described in detail later.

(54) The piston **52** is a cylindrical member having a bottom and is disposed within the cylinder **42** (the rear half of the spindle **40**) to be slidable along the driving axis DX. A rear end portion **571** of the piston **57** is operably coupled to the oscillating arm **551** of the oscillating member **55** via a coupling pin **573**. The piston **57** is configured to be reciprocated in the front-rear direction while the oscillating arm **551** of the oscillating member **55** is oscillated.

(55) The inner housing **20** of this embodiment is configured to house a portion of the motion converting mechanism **51** of the hammer mechanism **5**. More specifically, the inner housing **20** is configured such that the most part of the rotary member **54** and the oscillating member **55** that are disposed on the intermediate shaft **52** is accommodated in the internal space **200** of the inner housing **20**. In other words, the most part of the rotary member **54** and the oscillating member **55** is surrounded by the peripheral wall part **202** and the rear wall part **203** of the inner housing **20**.

(56) As shown in FIG. 3, the hammer element **58** is configured to linearly move in response to the reciprocation of the piston **57** and to strike the tool accessory **91**, so as to linearly drive the tool accessory **91** along the driving axis DX. In this embodiment, the hammer element **58** includes a striker **581** and an impact bolt **583**. The striker **581** is disposed within the piston **57** to be slidable along the driving axis DX. An internal space of the piston **57** behind the striker **581** (a space between the striker **581** and the bottom of the piston **57**) is configured as an air chamber **59** that serves as an air spring. The impact bolt **583** is disposed in front of the striker **581** within the tool holder **41** to be slidable along the driving axis DX. A space in front of the striker **581** communicates with the first space **101** via at least one through hole formed in the tool holder **41**.

(57) When the piston **57** reciprocates within the cylinder **42**, the pressure of the air in the air chamber **59** fluctuates. Consequently, the striker **581** slides within the piston **57** in the front-rear direction, owing to the effect of the air spring. More specifically, when the piston **57** is moved forward, the striker **581** is pushed forward at high speed, owing to the effect of the air spring, so as to strike the impact bolt **583**. The impact bolt **583** transmits the kinetic energy of the striker **581** to

the tool accessory **91**. Thus, the tool accessory **91** is linearly driven forward along the driving axis DX. When the piston **57** is moved rearward, the striker **581** is pulled rearward. Since the tool accessory **91** is pressed against a workpiece, the tool accessory **91** moves rearward together with the impact bolt **583**. In this manner, the hammer action is repeated by the hammer mechanism **5**. (58) As shown in FIG. **3**, the rotation transmitting mechanism **6** is operably coupled to the motor shaft **315** and is configured to transmit the rotation of the motor shaft **315** to the tool holder **41** (the spindle **40**). The rotation transmitting mechanism **6** includes a driving gear **61** that is disposed on the front end portion of the intermediate shaft **52**, and a driven gear **63** that is fixed around the cylinder **42** of the spindle **40** so as to mesh with the driving gear **61**. The driving gear **61** and the driven gear **63** form a gear speed reducer. A state of the driving gear **61** is changed between a first state and a second state by the above-described mode changing mechanism. In the first state, the driving gear **61** rotates integrally with the intermediate shaft **52**. In the second state, the driving gear **61** idly rotates relative to the intermediate shaft **52**. The driving gear **61** rotates integrally with the intermediate shaft **52** in the first state only when the action mode for performing the rotary action is selected.

(59) When the driving gear **61** is rotated integrally with the intermediate shaft **52** in response to the rotation of the motor shaft **315**, the spindle **40** is rotated integrally with the driven gear **63**, and thereby the tool accessory **91** that is held by the tool holder **41** is rotationally driven around the driving axis DX. In this manner, the rotary action is performed by the rotation transmitting mechanism **6**.

(60) The counterweight **7** and the supporting structure of the counterweight **7** are now described.

(61) As shown in FIGS. **6** and **7**, the counterweight **7** is supported by the support body **22** to be pivotable (to be oscillated) in the first space **101**, which is surrounded by the first housing part **11** and the inner housing **20**. As described above, the support body **22** is a member (part, component) that is originally separate (discrete) from the inner housing **20** and is coupled to the front end portion of the upper half of the inner housing **20**.

(62) More specifically, as shown in FIGS. **8** and **9**, the counterweight **7** of this embodiment is an annular (ring-like) member that is elongate in the up-down direction. The counterweight **7** is made of metal. The support body **22** includes an annular main body **221** and two arm parts **223** each protruding rearward from an upper portion of the main body **221**.

(63) The main body **221** supports the rear end portion of the spindle **40** (the cylinder **42**) via the bearing **402**, as described above (see FIG. **6**). The main body **221** is fixed to the front end portion of the inner housing **20** using screws, such that a contact surface that is defined at a rear end of the main body **221** is in contact with a contact surface that is defined on the front end portion of the inner housing **20**. The main body **221** is above the rotary member **54**, which is disposed on the intermediate shaft **52**, in the up-down direction.

(64) Each of the two arm parts **223** extends from a portion of the main body **221** above the rear end portion **571** of the piston **57** (i.e., a portion coupled to the oscillating arm **551**) into the internal space **200** of the inner housing **20**. The arm parts **223** are spaced apart from each other in the left-right direction and at substantially the same location in the up-down direction. The arm parts **23** extend in the front-rear direction to be in substantially parallel to each other. A rear end of each of the arm parts **223** is directly above the rotary member **54** and the ring part **550** of the oscillating member **55**.

(65) As shown in FIGS. **6** and **7**, a support hole **71** penetrates (extends) in the left-right direction through an upper end portion of the counterweight **7**. A support shaft **225** is inserted into the support hole **71**. The rear end portions of the two arm parts **223** of the support body **22** support opposite axial end portions of the support shaft **225**, respectively. More specifically, a support hole **224** penetrates (extends) in the left-right direction through the rear end portion of each of the arm parts **223**. The axial end portions of the support shaft **225** are respectively inserted into the support holes **224** of the two arm parts **223** and are thus supported by the two arm parts **223**. The

counterweight **7** is pivotably supported by the support body **22** via the support shaft **225** in this manner. The pivot axis PX of the counterweight **7** extends in the left-right direction above the rear end portion **571** of the piston **57**.

(66) An inner region of the ring part of the counterweight **7** includes an upper half and a lower half in the up-down direction. The upper half of the inner region is located at a position corresponding to the rear end portion **571** of the piston **57**. The lower half of the inner region is located at a position corresponding to the ring part **550** of the oscillating member **55**. The counterweight **7** is designed such that the rear end portion **571** of the piston **57** and the ring part **550** of the oscillating member **55** are movable within the inner region without interfering with the counterweight **7** during the hammer action.

(67) Further, as shown in FIGS. **6** and **9**, the counterweight **7** is operably coupled to the oscillating member **55**. More specifically, an engagement hole **73** is formed in the lower end portion of the counterweight **7**. The protrusion **552** of the oscillating member **55** is loosely inserted into the engagement hole **73**. Thus, when the oscillating member **55** is oscillated in response to the rotation of the intermediate shaft **52**, the counterweight **7** is driven by the protrusion **552** to be pivoted (oscillated) around the pivot axis PX. As described above, the protrusion **552** is coaxial with the oscillating arm **551** and protrudes in the opposite direction from the oscillating arm **551**. Therefore, the lower end portion of the counterweight **7** moves opposite to the piston **57** and the hammer element **58**. Thus, the counterweight **7** can reduce periodic vibration in the front-rear direction that is caused during the hammer action.

(68) As described above, in this embodiment, the counterweight **7** is supported, not by the inner housing **20**, but by the support body **22**, which is a member that is originally separate (discrete) from the inner housing **20**. Thus, the inner housing **20** can be optimally designed, without any constraints of the supporting structure of the counterweight **7**. For example, compared to a known structure in which a counterweight is directly supported by an inner housing itself, the shape of the inner housing **20** can be simplified. Further, the arm parts **223** can be easily placed in the support body **22**, which is advantageous in processing cost of the supporting structure. In this embodiment, the support body **22** not only supports the counterweight **7**, but also supports the rear end portion of the spindle **40**. Thus, a rational supporting structure of the counterweight **7** and the spindle **40** can be achieved without increasing the number of parts or components.

(69) Further, in the above-described known structure, a part or component (a support part) that supports the counterweight via a support shaft protrudes frontward from a peripheral wall part of the inner housing. This is because the support shaft needs to be inserted into a support hole or support holes of the supporting part in assembling. The support part and counterweight, as well as the driving mechanism and lubricant, are disposed in a space sealed by a seal member mounted around the peripheral wall part of the inner housing. However, since the counterweight needs to be driven by an oscillating member, the support shaft cannot be spaced apart frontward from the oscillating member beyond a certain limit. Thus, in the above-described known structure, there is a limit to how far the peripheral wall part of the inner housing can be extended frontward so that the seal member is positioned further frontward.

(70) In contrast, in this embodiment, the support body **22** supports the support shaft **225** in the internal space **200** of the inner housing **20**. Accordingly, there is no such a limit, and thus the inner housing **20** can be extended further frontward, compared to the known structure and the seal member **28** can be disposed in the front end portion of the inner housing **20**. Consequently, the air that flows within the second space **102** along the above-described air passage can effectively cool a wider range of the inner housing **20**.

(71) A structure for adjusting the pressure of air in the first space **101** is now described.

(72) As described above, the lubricant is in the first space **101** that is defined by the inner housing **20** and the first housing part **11**. The first space **101** is thus configured as a closed space, in order to restrict a leakage of the lubricant. On the other hand, if the first space **101** is made to be an air-tight

space, the hammer element **58** may malfunction (may not operate properly). Specifically, when heat is generated due to driving of the hammer mechanism **5**, the temperature and the pressure of the air in the first space **101** increase, and the pressure of the air in the space in front of the striker **581** within the spindle **40** also increases. Such a situation may lead to an imbalance between the pressure in the space in front of the striker **581** and the pressure in the air chamber **59**, which serves as an air spring, and thus the striker **581** may fail to properly operate in linear motion.

(73) In order to prevent such malfunction, the rotary hammer **1** of this embodiment has a structure that is configured to adjust the pressure in the first space **101** by allowing the air in the first space **101** to flow out of the first space **101** and flows into the second space **102**. Specifically, as shown in FIG. **10**, the inner housing **20** has an air ventilation hole (air ventilation passage) **24** that communicates with an inside and an outside of the first space **101**. However, it is not favorable that the lubricant leaks out of the first space **101** through the air ventilation hole **24**. For this reason, in this embodiment, the air ventilation hole **24** and a vicinity of the air ventilation hole **24** are configured to restrict a leakage of the lubricant.

(74) The air ventilation hole **24** is formed in the rear wall part **203** of the inner housing **20** at a position corresponding to a central portion of the second bevel gear **53** that is fixed to the intermediate shaft **52**. More specifically, a protrusion **205** protrudes frontward from a portion of the rear wall part **203** of the inner housing **20** that corresponds to (faces) a region that is radially inward of the inner periphery (the radially inner edge) of the gear teeth **531** of the second bevel gear **53**. The air ventilation hole **24** is a through hole that linearly penetrates (extends) through the protrusion **205** in the front-rear direction along the rotational axis RX of the second bevel gear **53**. The air ventilation hole **24** includes an inlet opening **241**, which faces the inside of the inner housing **20**, and an outlet opening **242**, which faces the outside of the inner housing **20**.

(75) A recess (a hole with a bottom) **25** that is configured to receive the protrusion **205** of the inner housing **20** is formed in the second bevel gear **53** and the intermediate shaft **52**. As described above, the second bevel gear **53** is fixed onto the outer periphery of the intermediate shaft **52** such that the rear end portion of the second bevel gear **53** protrudes rearward from the rear end of the intermediate shaft **52**. The recess **25** has an opening **251** on a rear end surface of the second bevel gear **53**, penetrates through the central portion of the teeth part of the second bevel gear **53**, and extends frontward into the central portion of the intermediate shaft **52**. The diameter of the recess **25** is larger than the diameter of the protrusion **205**. The protrusion **205** is inserted into the recess **25** through the opening **251** with a gap between an outer surface of the protrusion **205** and a surface that defines the recess **25** (i.e., such that the protrusion **205** is spaced apart from the surface that defines the recess **25**). A distal end of the protrusion **205** and thus the inlet opening **241** of the air ventilation hole **24** is spaced apart rearward from a bottom surface of the recess **25**. The air ventilation hole **24** communicates with an internal space of the recess **25**.

(76) Such a structure defines an air passage that extends from the opening **251**, passes through the inside of the recess **25**, and reaches the outside of the inner housing **20** by way of the air ventilation hole **24**. Thus, this air passage is configured such that the air needs to turn (i.e., a direction of an air flow is reversed) in the middle of the air passage.

(77) The recess **25** of this embodiment is formed by a through hole that penetrates (extends through) the central portion of the teeth part of the second bevel gear **53** and a recess (a hole with a bottom) that communicates with the through hole of the second bevel gear **53** and that is recessed frontward from the rear end surface of the intermediate shaft **52**. The diameter of the through hole of the second bevel gear **53** is larger than the diameter of the recess of the intermediate shaft **52**, so that the protrusion **205** can be easily inserted into the recess **25** in assembling.

(78) As shown in FIGS. **4** and **10**, a tubular part **207** is provided on the outer surface of the rear wall part **203** of the inner housing **20**. The tubular part **207** encircles the air ventilation hole **24**. A filter **208** is fitted into the tubular part **207**. The filter **208** is formed by any member that allows air to pass therethrough while absorbs and retains the lubricant therein. For example, felt or sponge

may be employed for the filter **208**.

(79) A portion of the rear wall part **203** of the inner housing **20** that faces the gear teeth **531** of the second bevel gear **53** has a shape that conforms to the shape of the teeth part. A distance between an inner surface of the portion of the inner housing **20** that faces the gear teeth **531** and tips of the gear teeth **531** is smaller than a tooth depth of the gear teeth **531** (a distance between a tooth tip and a tooth root at an outer peripheral edge of the gear teeth **531**). Thus, the distance between the tips of the gear teeth **531** and the inner housing **20** is extremely small. This configuration can limit an amount of the lubricant that enters the gap between the tips of the gear teeth **531** and the inner housing **20**. Further, the distance between the tips of the gear teeth **531** and the inner housing **20** is substantially uniform in the radial direction of the second bevel gear **53**.

(80) Similarly, a distance between an inner surface of a portion of the inner housing **20** that faces the gear teeth **331** of the first bevel gear **33** and tips of the gear teeth **331** is smaller than a tooth depth of the gear teeth **331** (a distance between a tooth tip and a tooth root at an outer peripheral edge of the gear teeth **331**). Thus, the distance between the tips of the gear teeth **331** and the inner housing **20** is extremely small. Further, the distance between the tips of the gear teeth **331** and the inner housing **20** is substantially uniform in the radial direction of the first bevel gear **33**.

(81) In this embodiment, the bearing **522** that supports the rear end portion of the intermediate shaft **52** is supported, not directly by the rear wall part **203**, but by the bearing support **23** that is fixed to the inner housing **20** within the inner housing **20**. Therefore, as described above, the rear wall part **203** of the inner housing **20** can be designed to have an appropriate shape such that the lubricant cannot easily pass through the air passage.

(82) When the hammer mechanism **5** generates heat during the driving of the hammer mechanism **5**, the air in the first space **101** expands to increase the pressure in the first space **101**. When this happens, the air in the first space **101** flows out through the air ventilation hole **24** into the second space **102**, in which the pressure is lower. Specifically, the air enters the bearing support **23** (a space between the bearing **522** and the rear wall part **203**) mainly through the openings **230** from the outside of the bearing support **23**. This air passes through the gap between the second bevel gear **53** and the inner surface of the inner housing **20** and the gap between the first bevel gear **33** and the inner surface of the inner housing **20**, and reaches the opening **251** of the recess **25**. The air that flows into the recess **25** via the opening **251** flows frontward through the gap between the outer surface of the protrusion **205** and the surface that defines the recess **25**, flows into the air ventilation hole **24** via the inlet opening **241**, and flows out into the second space **102** via the outlet opening **242** and the filter **208**.

(83) On the other hand, a possibility of leakage of the lubricant from the first space **101** into the second space **102** can be effectively reduced. Specifically, during the driving of the hammer mechanism **5**, the second bevel gear **53** is rotated, so that the lubricant between the gear teeth **531** of the second bevel gear **53** and the inner surface of the inner housing **20** is delivered (fed) radially outward by the centrifugal force. The lubricant can flow out of the bearing support **23** through the openings **203** of the bearing support **23**. In this manner, the second bevel gear **53**, which is a part of the driving mechanism **4**, is utilized to reduce a possibility that the lubricant reaches the opening **251** of the recess **25** that is provided at the central portion of the second bevel gear **53** to communicate with the air ventilation hole **24**. The second bevel gear **53** has a larger diameter than the first bevel gear **33**. Therefore, a passage from the outer peripheral edge (the radially outer edge) to the central portion of the gear teeth **531** can be made longer, compared to a passage from an outer peripheral edge (the radially outer edge) to a central portion of the gear teeth **331**. Further, the recess **25** that receives the protrusion **205** can be formed easily. Thus, there is an advantage to utilize the second bevel gear **53** to form the effective air passage.

(84) In addition, as described above, the passage from the opening **251** of the recess **25** to the outlet opening **242** of the air ventilation hole **24** first extends frontward and then turns to extend rearward. This configuration can further reduce the possibility of the leakage of the lubricant. Further, even if

the lubricant leaks out from the outlet opening **242** of the air ventilation hole **24**, the filter **208** that is fitted in the tubular part **207** absorbs and retains the lubricant, which reduces a possibility that the lubricant spreads in the second space **102**.

(85) The above-described embodiment is merely exemplary, and the reciprocating tool according to the present disclosure is not limited to the rotary hammer **1** of the above-described embodiment. For example, the following non-limiting modifications may be made. Further, at least one of these modifications may be employed in combination with at least one of the rotary hammer **1** of the above-described embodiment and the claimed features.

(86) For example, the reciprocating tool according to the present disclosure may be embodied as a power tool having a hammer mechanism that is configured to perform only the hammer action (e.g., an electric hammer (a demolition hammer or a scraper)), or as a reciprocating cutting tool that is configured to reciprocate a tool accessory (for example, a blade) to perform a cutting operation.

(87) The structures and/or arrangements of the body housing, the inner housing, the motor, and the driving mechanism for the tool accessory in the reciprocating tool according to the present disclosure may be appropriately changed from those in the above-described embodiment. For example, the motor may be disposed such that the motor axis extends in parallel to the driving axis.

(88) The counterweight according to the present disclosure need not necessarily have an annular shape. For example, the counterweight may have a U-shape. The counterweight of this modification may be supported by the support body **22** via the support shaft **225** that is inserted into support holes formed in two end portions of the U-shaped counterweight. In another modification, a support shaft may be integrally formed with the counterweight and may be pivotably supported by the support body **22**. The counterweight and the oscillating member of the motion converting mechanism according to the present disclosure may be operably coupled to each other at a position different from that of the above-described embodiment.

(89) The support body according to the present disclosure need not necessarily support the tool accessory holding member (the spindle **40**). Instead, the support body may only serve to support the counterweight. The shape of the support body (e.g., the shape of the main body **221** and/or the shape and/or the number of the arm parts **223**) may be appropriately changed. Further, the support body may be attached (fixed) to the inner housing in a state in which an entirety of the support body is within the inner housing.

(90) Further, in view of the nature of the present invention, the above-described embodiment, and the modifications thereof, the following Aspects A1 to A8 can be provided. Any one or more of the following Aspects A1 to A8 can be employed in combination with any one or more of the above-described embodiments, the above-described modifications, and the claimed features.

Aspect A1

(91) The inner housing is fitted into a portion of the body housing from the rear side.

Aspect A2

(92) The inner housing is a hollow body that has an open front end, a closed circumference around the driving axis, and a closed rear end.

Aspect A3

(93) The support shaft is between a front end and a rear end of the inner housing in the front-rear direction.

Aspect A4

(94) An entirety of the counterweight is between a front end and a rear end of the inner housing in the front-rear direction (i.e., an entirety of the counterweight is within the inner housing).

Aspect A5

(95) A portion of the support body is forward of the inner housing in the front-rear direction.

Aspect A6

(96) At least a portion of the inner housing is forward of the air inlet opening in the front-rear direction, the body housing includes a guide rib that is configured to guide the air that flows into

the body housing through the air inlet opening, and the guide rib is adjacent to the air inlet opening and is at least partially inclined frontward as the guide rib extends away from the air inlet opening.

Aspect A7

(97) The motion converting mechanism includes: a rotary member that is disposed around the intermediate shaft and is configured to rotate integrally with the intermediate shaft, an oscillating member that is operably coupled to the rotary member and is configured to oscillate in an extension direction of the driving axis in response to rotation of the rotary member, and a piston that is operably coupled to the oscillating member and is configured to reciprocate along the driving axis within the tool accessory holding member.

Aspect A8

(98) The motion converting mechanism includes: a rotary member that is disposed around the intermediate shaft and is configured to rotate integrally with the intermediate shaft, and an oscillating member that is operably coupled to the rotary member and is configured to oscillate in an extension direction of the driving axis in response to rotation of the rotary member, wherein the counterweight is operably coupled to the oscillating member.

(99) The following Aspects B1 to B13 can be provided for another non-limiting object to provide a power tool having an improved structure for ventilating air from a housing space of a hammer (striking) mechanism that utilizes an air spring. Any one of the following Aspects B1 to B13 can be employed alone or two or more of them can be employed in combination with each other.

Alternatively, any one or more of the following Aspects B1 to B13 can be employed in combination with any one of the rotary hammer **1** of the above-described embodiment, the above-described modifications, the above-described Aspects A1 to A8, and the claimed features.

Aspect B1

(100) A power tool comprising: a motor that has a motor shaft to which a first bevel gear is fixed; a hammer mechanism that (i) is operably coupled to the motor shaft via a second bevel gear that meshes with the first bevel gear, and (ii) is configured to linearly drive a tool accessory along a driving axis using an action of an air spring; a body housing that houses the motor and the hammer mechanism; and an inner housing that is within the body housing, wherein: a closed housing space is defined within the body housing, the hammer mechanism and lubricant are in the housing space, the inner housing defines at least a portion of the housing space, the inner housing has an air ventilation hole (air ventilation passage), the air ventilation hole has an inlet opening that faces an inside of the inner housing and an outlet opening that faces an outside of the inner housing, and the inlet opening of the air ventilation hole is radially inward of a radially inner edge (an inner peripheral edge) of gear teeth of a first one of the first bevel gear and the second bevel gear.

(101) In the power tool of this aspect, the hammer mechanism and the lubricant are accommodated in the housing space that is defined within the body housing. The housing space is a closed space that is at least partially defined by the inner housing, and air is allowed to flow in and out of the housing space through the air ventilation hole of the inner housing. Owing to this structure, when the internal pressure of the housing space is increased due to heat generation caused by driving of the hammer mechanism, the air flows out of the housing space through the air ventilation hole. Consequently, the internal pressure of the housing space is adjusted and a possibility that the air spring does not properly operate can be reduced.

(102) Further, the inlet opening of the air ventilation hole is radially inward of (at the radially inner side of) the radially inner edge of the gear teeth of the first one of the first bevel gear and the second bevel gear. Owing to this structure, when the hammer mechanism is driven and thus the first one of the first bevel gear and the second bevel gear is rotated, the lubricant cannot easily reach the inlet opening of the air ventilation hole. The first bevel gear and the second bevel gear are components that are used to transmit power from the motor to the hammer mechanism. Thus, by utilizing the first bevel gear or the second bevel gear, the structure of this aspect can effectively reduce the possibility of leakage of the lubricant without increasing the number of components.

Aspect B2

(103) The power tool as defined in Aspect B1, wherein the inlet opening of the air ventilation hole is radially inward of the radially inner edge of the gear teeth of the second bevel gear.

(104) According to this aspect, the air ventilation hole is relatively easily formed in the inner housing. Generally, in order to reduce rotation speed, the diameter of the second bevel gear is larger than the diameter of the first bevel gear. Thus, in such a case, by providing the air ventilation hole that corresponds to the second bevel gear, a passage from the radially outside of the second bevel gear to the inlet opening of the air ventilation hole can be made longer than a structure in which the air ventilation hole corresponds to the first bevel gear. Therefore, in a case in which the second bevel gear has a larger diameter than the first bevel gear, it is preferable that the position of the air ventilation hole is determined, corresponding to the second bevel gear.

Aspect B3

(105) The power tool as defined in Aspect B2, wherein a rotational axis of the second bevel gear extends through the inlet opening of the air ventilation hole.

(106) According to this aspect, the inlet opening can be disposed at a position where it is most difficult for the lubricant to reach.

Aspect B4

(107) The power tool as defined in Aspect B2 or B3, further comprising: an intermediate shaft that is operably coupled to the motor shaft and is rotatable around a rotational axis that extends in parallel to the driving axis, wherein: the hammer mechanism includes a motion converting mechanism that (i) is partially disposed on the intermediate shaft and (ii) is configured to convert rotation of the intermediate shaft into linear motion in an extension direction of the driving axis for driving the tool accessory, and the second bevel gear is fixed to the intermediate shaft.

(108) According to this aspect, the possibility of leakage of the lubricant can be effectively reduced, utilizing the second bevel gear fixed to the intermediate shaft on which a portion of the motion converting mechanism is disposed.

Aspect B5

(109) The power tool as defined in Aspect B4, further comprising: a bearing that is supported by the inner housing and rotatably supports the second bevel gear, wherein the intermediate shaft is rotatably supported via the second bevel gear.

(110) According to this aspect, the intermediate shaft can be made shorter, compared to a structure in which a portion of the intermediate shaft is directly supported by the bearing.

Aspect B6

(111) The power tool as defined in any one of Aspects B1 to B5, wherein: the inner housing has a protrusion that protrudes toward the first one of the first bevel gear and the second bevel gear, the air ventilation hole penetrates the protrusion, the inlet opening is at a distal end of the protrusion, and the protrusion extends to at least an inside of the first one of the first bevel gear and the second bevel gear in an axial direction of the first one of the first bevel gear and the second bevel gear.

(112) According to this Aspect, a passage is formed that extends from an axial end of the first one of the first bevel gear and the second bevel gear to the inlet opening of the air ventilation hole. Therefore, the air that has passed the passage needs to turn (i.e., needs to change its proceeding direction) to enter the air ventilation hole. Accordingly, the possibility of leakage of the lubricant can be further reliably reduced.

Aspect B7

(113) The power tool as defined in Aspect B6, further comprising: a rotary shaft that is rotatable around an axis that extends in the front-rear direction of the power tool, wherein: the second bevel gear is fixed to a rear end portion of the rotary shaft, a rear end of the second bevel gear is rearward of a rear end of the rotary shaft, and the protrusion protrudes frontward from the inner housing into an inside of the rotary shaft.

(114) According to this aspect, the passage penetrates through the second bevel gear, extends into

the inside of the rotary shaft and to the inlet opening of the air ventilation hole. Thus, for example, a portion of the passage in the second bevel gear and a portion of the passage in the rotary shaft can be designed, independent of each other.

Aspect B8

(115) The power tool as defined in any one of Aspects B1 to B7, wherein a distance between an inner surface of the inner housing and tips of the gear teeth of the first one of the first bevel gear and the second bevel gear is smaller than a tooth depth of the gear teeth.

Aspect B9

(116) The power tool as defined in Aspect B8, wherein the distance between the inner surface of the inner housing and the tips of the gear teeth of the first one of the first bevel gear and the second bevel gear is substantially uniform.

(117) These aspects can make it more difficult for the lubricant to pass through between the inner surface of the inner housing and the tips of the gear teeth of the first one of the first bevel gear and the second bevel gear.

Aspect B10

(118) The second bevel gear has a larger diameter than the first bevel gear.

Aspect B11

(119) The first bevel gear is disposed in the housing space.

Aspect B12

(120) At least one of the second bevel gear and the intermediate shaft has a recess that extends along the rotational axis of the intermediate shaft, the protrusion of the inner housing protrudes into the recess, and the air ventilation hole communicates with an internal space of the recess.

Aspect B13

(121) The bearing is supported by a bearing support that is originally separate from the inner housing and is attached to the inner housing within the housing space, and the second bevel gear is between the inner housing and the bearing support.

(122) The above-described embodiment is merely exemplary, and the power tool according to the Aspects B1 to B13 of the present disclosure is not limited to the rotary hammer **1** of the above-described embodiment. For example, the following non-limiting modifications may be made. Further, at least one of these modifications may be employed in combination with at least one of the rotary hammer **1** of the above-described embodiment, the above-described modifications, the above-described Aspects A1 to A8, the above-described Aspects B1 to B13, and the claimed features.

(123) For example, the power tool according to the present disclosure may be embodied as a power tool having a hammer (striking) mechanism that is configured to perform only the hammer action (e.g., an electric hammer (a demolition hammer or a scraper)). The structures and/or arrangements of the body housing, the inner housing, the motor, and the hammer mechanism in the power tool may be appropriately changed from those in the above-described embodiment.

(124) In the above-described embodiment, the inlet opening **241** of the air ventilation hole **24** is positioned to correspond to the center (the rotational axis RX) of the second bevel gear **53** that is fixed to the intermediate shaft **52**, and thus it is preferable in that the lubricant is the most difficult to reach the inlet opening **241**. However, the position of the inlet opening **241** may be changed to any position, as long as it corresponds to a region that is radially inward of the radially inner edge of the gear teeth **531** of the second bevel gear **53**. Further, the air ventilation hole **24** may extend along a straight line that is oblique relative to the rotational axis RX. Alternatively, the air ventilation hole **24** may be bent or curved. Further, the protrusion **205** of the inner housing **20** and the recess **25** of the second bevel gear **53** and the intermediate shaft **52** may be omitted, and the inlet opening **241** of the air ventilation hole **24** may directly face the rear end surface of the second bevel gear **53**.

(125) In the above-described embodiment, the position of the air ventilation hole **24** corresponds to

the second bevel gear **53**, considering that the second bevel gear **53** has a larger diameter than the first bevel gear **33** and thus has the above-described advantages. However, the air ventilation hole **24** may be provided at a position that corresponds to the first bevel gear **33**.

DESCRIPTION OF THE REFERENCE NUMERALS

(126) **1**: rotary hammer, **10**: body housing, **101**: first space, **102**: second space, **105**: air inlet opening, **106**: guide rib, **107**: air outlet opening, **11**: first housing part, **111**: barrel part, **113**: opening, **15**: second housing part, **150**: rear wall part, **151**: upper half, **153**: lower half, **17**: handle, **171**: grip part, **173**: trigger, **175**: switch, **20**: inner housing, **200**: internal space, **201**: opening, **202**: peripheral wall part, **203**: rear wall part, **204**: groove, **205**: protrusion, **207**: tubular part, **208**: filter, **22**: support body, **221**: main part, **223**: arm part, **224**: support hole, **225**: support shaft, **23**: bearing support, **230**: opening, **24**: air ventilation hole, **241**: inlet opening, **242**: outlet opening, **25**: recess, **251**: opening, **28**: seal member, **30**: controller, **31**: motor, **310**: motor body, **315**: motor shaft, **321**: bearing, **323**: bearing, **33**: first bevel gear, **331**: gear teeth, **35**: fan, **4**: driving mechanism, **40**: spindle, **401**: bearing, **402**: bearing, **41**: tool holder, **410**: opening, **42**: cylinder, **5**: hammer mechanism, **51**: motion converting mechanism, **52**: intermediate shaft, **521**: bearing, **522**: bearing, **53**: second bevel gear, **531**: gear teeth, **533**: mount part, **54**: rotary member, **55**: oscillating member, **550**: ring part, **551**: oscillating arm, **552**: protrusion, **57**: piston, **571**: rear end portion, **573**: coupling pin, **58**: hammer element, **581**: striker, **583**: impact bolt, **59**: air chamber, **6**: rotation transmitting mechanism, **61**: driving gear, **63**: driven gear, **7**: counterweight, **71**: support hole, **73**: engagement hole, **91**: tool accessory, **93**: battery, DX: driving axis, MX: motor axis, PX: pivot axis, RX: rotational axis

Claims

1. A reciprocating tool that is configured to linearly reciprocate a tool accessory, the reciprocating tool comprising: a motor that has a motor shaft that is rotatable around a motor axis; a motion converting mechanism that is (i) operably coupled to the motor shaft and (ii) is configured to convert rotation into linear reciprocating motion along a driving axis that defines a front-rear direction of the reciprocating tool; a body housing; an inner housing that (i) is within the body housing and (ii) houses at least a portion of the motion converting mechanism; a support body that (i) is originally separate from the inner housing and (ii) is attached to the inner housing; and a counterweight that (i) is operably coupled to the motion converting mechanism and (ii) is configured to be driven by the motion converting mechanism, wherein: the counterweight is (i) supported by the support body to be pivotable around a pivot axis that extends in a direction orthogonal to the driving axis and (ii) supported by the support body via a support shaft that extends along the pivot axis; the support body directly supports the support shaft; and at least a portion of the counterweight is within the inner housing.
2. The reciprocating tool as defined in claim 1, wherein: the support shaft is in an internal space of the inner housing.
3. The reciprocating tool as defined in claim 2, wherein the support body includes: a main body that is at least partially on an outside of the inner housing, and an arm part that protrudes from the main body into the inner housing and supports the support shaft.
4. The reciprocating tool as defined in claim 3, wherein: the body housing includes (i) a first housing part that houses the motion converting mechanism, and (ii) a second housing part that is coupled to a rear end portion of the first housing part and houses the motor, the inner housing is a hollow body that has an open front end, a closed circumference around the driving axis, and a closed rear end, and the inner housing is fitted into the rear end portion of the first housing part.
5. The reciprocating tool as defined in claim 4, wherein: at least a portion of the main body of the support body is disposed frontward of the inner housing in the front-rear direction, and the arm part extends rearward from the main body into the inner housing.

6. The reciprocating tool as defined in claim 5, further comprising: a seal that is disposed between an outer surface of a front end portion of the inner housing and an inner surface of the rear end portion of the body housing, wherein: the seal is configured to partition an internal space of the body housing into (i) a first space in which the motion converting mechanism is disposed and (ii) a second space in which the motor is disposed, and the seal is disposed frontward of the pivot axis of the counterweight in the front-rear direction.
7. The reciprocating tool as defined in claim 6, wherein: an extension direction of the pivot axis defines a left-right direction of the reciprocating tool, a direction that is orthogonal to the front-rear direction and the left-right direction defines an up-down direction of the reciprocating tool, and the body housing has: an air inlet opening that (i) is above an upper end of the seal in the up-down direction and rearward of the seal in the front-rear direction and (ii) communicates with the second space, and an air outlet opening that (i) is below a lower end of the seal in the up-down direction and (ii) communicates with the second space.
8. The reciprocating tool as defined in claim 2, further comprising: a seal member that is disposed between the body housing and the inner housing, wherein the seal member is frontward of the pivot axis of the counterweight in the front-rear direction.
9. The reciprocating tool as defined in claim 8, wherein: an extension direction of the pivot axis defines a left-right direction of the reciprocating tool, a direction that is orthogonal to the front-rear direction and the left-right direction defines an up-down direction of the reciprocating tool, and the body housing has: an air inlet opening that is above an upper end of the seal member in the up-down direction, and an air outlet opening that is below a lower end of the seal member in the up-down direction.
10. The reciprocating tool as defined in claim 1, further comprising: a spindle that is tubular and configured to hold the tool accessory to be movable along the driving axis, and a striker that (i) is disposed within the spindle and (ii) is configured to be driven by the motion converting mechanism to apply a striking force to the tool accessory, wherein the support body supports the spindle.
11. The reciprocating tool as defined in claim 10, further comprising: an intermediate shaft that (i) is operably coupled to the motor shaft and (ii) is rotatable around a rotational axis that extends in parallel to the driving axis, wherein a portion of the motion converting mechanism is disposed on the intermediate shaft.
12. The reciprocating tool as defined in claim 11, wherein the motor axis intersects the rotational axis of the intermediate shaft.
13. The reciprocating tool as defined in claim 12, wherein: an extension direction of the pivot axis defines a left-right direction of the reciprocating tool, a direction that is orthogonal to the front-rear direction and the left-right direction defines an up-down direction of the reciprocating tool, the body housing includes an air inlet opening and an air outlet opening, and the air inlet opening is above the driving axis in the up-down direction.
14. The reciprocating tool as defined in claim 13, wherein: an air passage is defined within the body housing, and the air passage is configured such that air, which has flowed into the body housing through the air inlet opening, flows along the inner housing, passes through an inside of the motor, and flows out of the body housing through the air outlet opening.
15. The reciprocating tool as defined in claim 10, further comprising: a bearing that supports the spindle to be rotatable around the driving axis, wherein: the support body includes (i) a main body that is at least partially disposed frontward of the inner housing, and (ii) an arm part that protrudes rearward from the main body, the bearing is fitted in the main body of the support body, the arm part supports a support shaft that extends along the pivot axis within the inner housing, and the counterweight is supported by the support body via the support shaft.
16. A reciprocating tool that is configured to linearly reciprocate a tool accessory, the reciprocating tool comprising: a motor that has a motor shaft that is rotatable around a motor axis; a motion converting mechanism that is (i) operably coupled to the motor shaft and (ii) is configured to

convert rotation into linear reciprocating motion along a driving axis that defines a front-rear direction of the reciprocating tool; a body housing; an inner housing that (i) is within the body housing and (ii) houses at least a portion of the motion converting mechanism; a support body that (i) is originally separate from the inner housing and (ii) is attached to the inner housing; and a counterweight that (i) is operably coupled to the motion converting mechanism and (ii) is configured to be driven by the motion converting mechanism, wherein: the counterweight is supported by the support body to be pivotable around a pivot axis that extends in a direction orthogonal to the driving axis; at least a portion of the counterweight is within the inner housing, the counterweight is supported by the support body via a support shaft that extends along the pivot axis, the support shaft is in an internal space of the inner housing, the support body includes: a main body that is at least partially on an outside of the inner housing, and an arm part that protrudes from the main body into the inner housing and supports the support shaft, the body housing includes (i) a first housing part that houses the motion converting mechanism, and (ii) a second housing part that is coupled to a rear end portion of the first housing part and houses the motor, the inner housing is a hollow body that has an open front end, a closed circumference around the driving axis, and a closed rear end, the inner housing is fitted into the rear end portion of the first housing part, at least a portion of the main body of the support body is disposed frontward of the inner housing in the front-rear direction, the arm part extends rearward from the main body into the inner housing, wherein the reciprocating tool further comprises a seal that is disposed between an outer surface of a front end portion of the inner housing and an inner surface of the rear end portion of the body housing, wherein: the seal is configured to partition an internal space of the body housing into (i) a first space in which the motion converting mechanism is disposed and (ii) a second space in which the motor is disposed, and the seal is disposed frontward of the pivot axis of the counterweight in the front-rear direction.
